

VIGNA DI VALLE, Italy

WMO: 162240

Lat: 42.0803N

Lon: 12.2114E

Elev: 266

StdP: 98.17

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	1.9	-7.5	2.1	3.8	-5.4	2.5	4.4	11.4	5.6	10.4	6.3	5.8	30	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	33.1	21.1	31.6	20.9	30.2	20.8	24.1	28.4	23.2	27.7	22.4	27.1	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.3	26.2	21.9	17.1	25.2	21.0	16.2	24.6	74.1	28.1	70.3	27.7	67.3	27.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.2	7.1	-0.9	35.3	2.0	1.9	-2.4	36.7	-3.6	37.7	-4.7	38.8	-6.2	40.1		
			-3.3	25.5	1.7	1.2	-4.5	26.3	-5.4	27.0	-6.4	27.7	-7.6	28.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	7.7	8.2	10.5	13.9	17.3	22.1	25.0	24.9	21.1	17.1	12.5	8.9
	DBStd	6.70	2.57	2.87	2.70	2.47	2.77	3.05	2.18	2.44	2.41	2.64	2.86	2.64
	HDD10.0	231	77	61	27	1	0	0	0	0	0	1	11	53
	HDD18.3	1581	329	285	244	135	53	5	0	0	4	54	177	294
	CDD10.0	2345	7	9	42	117	227	363	466	462	332	219	84	18
	CDD18.3	653	0	0	0	1	22	118	207	204	86	14	1	0
	CDH23.3	4665	0	0	0	6	107	823	1722	1638	349	20	0	0
	CDH26.7	1612	0	0	0	0	13	247	652	634	66	1	0	0
Wind	WSAvg	3.0	3.1	3.5	3.3	2.8	2.8	2.7	2.8	2.7	2.8	2.8	3.0	3.2
Precipitation	PrecAvg	1009	120	102	81	75	55	44	18	51	94	107	144	127
	PrecMax	1360	222	200	214	222	159	126	53	141	352	213	462	338
	PrecMin	655	9	5	16	22	3	8	0	0	0	0	30	29
	PrecStd	181	62	58	50	60	37	33	17	41	82	69	103	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	16.2	20.0	24.1	28.3	32.7	34.6	35.7	30.6	25.2	20.2	16.0
		MCWB	11.8	10.7	11.3	15.8	19.5	21.2	21.8	21.1	20.2	17.6	16.7	13.7
	2%	DB	13.4	14.3	17.7	22.0	26.1	31.1	33.0	33.7	28.5	23.6	18.7	14.5
		MCWB	11.2	10.3	12.1	15.2	18.2	20.8	21.0	21.3	20.4	18.0	15.9	12.3
	5%	DB	12.4	13.0	16.0	20.2	24.3	29.6	31.6	31.8	26.9	22.1	17.4	13.6
		MCWB	10.4	10.0	11.9	14.4	17.5	20.7	21.1	20.9	19.7	17.6	15.1	11.6
	10%	DB	11.4	12.0	14.6	18.5	22.6	27.9	30.2	30.1	25.2	20.9	16.2	12.7
		MCWB	9.5	9.9	11.1	13.6	16.9	20.0	20.9	20.7	19.1	17.2	14.3	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.6	14.6	17.7	21.0	23.8	25.6	25.2	23.9	20.3	18.2	14.6
		MCDB	13.8	14.3	16.7	21.7	25.6	28.4	29.5	28.7	26.2	22.1	19.5	15.4
	2%	WB	11.8	11.7	13.4	16.2	19.6	22.7	24.4	23.9	22.2	19.4	16.6	13.3
		MCDB	12.9	12.8	15.7	20.4	24.1	27.7	28.7	28.4	25.5	21.4	17.9	14.0
	5%	WB	10.8	10.9	12.5	15.1	18.5	21.8	23.4	23.0	21.1	18.6	15.6	12.3
		MCDB	11.9	12.4	14.7	19.0	22.8	27.0	27.9	27.7	24.6	20.8	16.9	13.1
	10%	WB	9.8	10.0	11.6	14.2	17.5	20.9	22.5	22.2	20.3	17.7	14.5	11.2
		MCDB	11.1	11.5	13.7	17.5	21.4	25.8	27.2	27.0	23.7	20.1	15.9	12.3
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.9	6.8	7.7	8.1	8.7	9.3	9.2	7.8	6.6	5.3	4.9
		MCDBR	6.0	7.2	8.9	9.8	10.2	10.4	11.0	11.1	9.2	7.9	6.0	5.5
		MCWBR	4.7	5.1	5.6	5.8	5.4	4.8	4.7	4.6	4.5	4.6	4.3	4.2
	5% WB	MCDBR	5.1	6.0	7.6	8.7	9.3	9.5	9.5	9.2	8.0	6.9	5.4	4.6
		MCWBR	4.7	4.8	5.4	5.8	5.7	5.2	5.3	5.1	5.0	4.9	4.5	4.2
Clear-Sky Solar Irradiance	taub	0.330	0.350	0.385	0.415	0.420	0.414	0.406	0.406	0.397	0.384	0.361	0.331	
	taud	2.469	2.395	2.298	2.226	2.254	2.315	2.339	2.352	2.373	2.394	2.397	2.468	
	Ebn at Noon	812	849	856	854	857	861	864	853	834	798	761	776	
	Edh at Noon	76	95	117	135	135	127	123	118	108	94	81	70	
All-Sky Solar Radiation	RadAvg	1.75	2.66	3.88	5.17	6.32	7.19	7.38	6.44	4.72	3.20	1.96	1.56	
	RadStd	0.17	0.27	0.45	0.38	0.54	0.32	0.34	0.34	0.23	0.28	0.18	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	+0.42	N/A	N/A	+0.59	N/A	N/A	N/A	-88	+140	+69

Nomenclature: See separate page